



Research Support Assistant – Longitudinal Executive and Attentional Functions (LEAF) Study

1 JOB DESCRIPTION

Faculty / School or Division: School of Psychological Science

Faculty/School or Division Address: 12a Priory Road, Bristol BS8 1TU

Job Family:	Research & Teaching		
Grade:	H	Salary range:	£32,982-£37,099 per annum
Hours of work:	35 hours per week	Contract type:	Open ended with funding for 2 years (31/08/2026)
Work pattern:	Full time	Vacancy Reference Number: ACAD107646	

1.1 Main Job Purpose

The Research Support Assistant will provide closely defined research support for the Longitudinal Executive and Attentional Functions (LEAF) study through the application of a range of techniques and will collect and analyse data that enable the project to progress. The role holder will seek to undertake training and development in order to gain the necessary skills and experience to apply for future Research Assistant positions, with support and guidance from line manager and project team.

1.2 Standard Responsibilities

Under supervision of the principal investigator, the post holder will:

- Take primary responsibility for the day-to-day running of test sessions, including (but not limited to):
 - Setting up for test sessions.
 - Running test sessions with other members of the team (two testers run the sessions together).
 - Post-session tasks: tidying up the lab, backing up data, putting buccal samples into the freezer, passing relevant forms and information on to the team's administrator.
- Have daily participant contact and maintain a high level of organisation and study management, always being on top of the schedule for each day and the tasks that need doing for the successful running of the study.

- Be proactive in resolving ('thinking on the feet') any technical issues or problems in data collection or management as they occur and reporting these to the principal investigator as soon as possible.
- Regularly back up the project data, including videos, to ensure all data has been collected adequately and is securely stored both on local drives and in the University-approved data archives.
- Code behavioural data from video recordings and train students and interns on agreed coding protocols in order to establish adequate inter-coder reliability (full training will be provided).
- Clean and process EEG, fNIRS and ECG data to agreed protocols, and provide training in these protocols when needed (full training will be provided).
- Regularly review testing protocols with the PI and her team to maintain a high level of task administration consistency and report any observed discrepancies.
- Implement agreed edits to testing, coding and analysis protocols.
- Assist the PI and wider research team with administrative tasks as needed (such as placing orders through the University's procurement system).
- Undertake such other duties as may be required from time to time that are commensurate with the grade and responsibilities of this post.

1.3 Relationships

Line manager: Karla Holmboe, Associate Professor in Developmental Science

Line manager to (where appropriate): N/A

1.4 Pre-employment Checks

The post holder will be subject to a satisfactory **enhanced with child barred list DBS check** as a requirement to operate within role. This check will be completed prior to taking up post, with rechecks required every 3 years while in role.

2 PERSON SPECIFICATION

2.1 Standard Skills, Knowledge & Experience Required

Criteria	Essential	Desirable	Assessed By A/I/T/O
First degree in a subject relevant to the field of work	Y		A
Demonstrable expertise in a range of research techniques and methods	Y		A, I
Ability to analyse data and present findings verbally/written	Y		A, I, T
Ability to work independently and seek guidance when unclear	Y		A, I
Ability to organise your own workload, consumables and other resources	Y		A, I
Understanding of health and safety requirements within a research environment	Y		A

Some previous experience in a research environment		Y	A, I
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2.2 Additional Skills, Knowledge & Experience (if applicable)

Criteria	Essential	Desirable	Assessed By A/I/T/O
Prior knowledge of and keen interest in experimental developmental psychology / developmental cognitive neuroscience	Y		A, I
Ability to, under supervision, acquire skills in novel and specialised hardware and software systems, including neuroimaging systems, and to report and trouble-shoot technical difficulties promptly and effectively	Y		A, I
Excellent organisational skills, time management, and attention to detail	Y		A, I, T
Ability to manage a large amount of data responsibly and securely	Y		A, I
Excellent communication skills (both verbal and written), including the ability to communicate effectively with other team members, colleagues in the School of Psychological Science, and participating families.	Y		A, I
Excellent interpersonal skills, including the ability to put parents and children at ease, and to respond sensitively to verbal and non-verbal cues from infants and their parents.	Y		A, I
Experience working with young children and their families.		Y	A
Experience with one or more of the specialised methodologies involved in the project: eye-tracking, video coding of behaviour, electroencephalography (EEG/ERP), functional near-infrared spectroscopy (fNIRS), electrocardiography (ECG), genetics.		Y	A
Demonstrable interest in the development of executive functions through previous work or reading/self-study.		Y	A
Evidence of showing exceptional initiative in pursuing psychological research as part of a degree and/or by undertaking additional research experience before, during or after a degree.		Y	A
Interest in pursuing a PhD in developmental psychology or developmental cognitive neuroscience.		Y	A

A – Application Form

I – Interview

T – Test or Scenario

O – Other e.g. presentation, focus group

3 OTHER RELEVANT INFORMATION

3.1 Further information about the project

Executive functions (EFs) are a set of cognitive abilities that allow us to guide our behaviour and make adaptive decisions. They provide us with the ability to control our thoughts and actions, solve problems, and multi-task. Without EFs we would be largely controlled by immediate circumstances and habitual responses.

EFs are particularly challenged in young children, who struggle to maintain and manipulate information in working memory, inhibit strong response tendencies and shift between different perspectives and actions. They are also compromised in children with developmental disorders and those with academic and social difficulties. Between 3 and 5 years of age EF skills improve rapidly. However, relatively little is known about EFs before this age.

The Longitudinal Executive and Attentional Functions (LEAF) study aims to push down the boundary for successful EF assessment by using a new task battery specifically designed for 9- to 30-month-old children. The battery has been created by adjusting existing tasks (e.g., to make them age-appropriate) and also by developing some entirely new tasks (e.g., Holmboe et al., 2021). In the LEAF study, children are assessed on the new battery alongside concurrent measures of brain (EEG and fNIRS) and heart rate activity (ECG). The potential impact of common genetic variation on early EF development is also investigated. The overarching aim of the LEAF study is to provide important knowledge on the very earliest development of executive functions, the biological substrate and correlates of this development, and individual trajectories over time. Outcomes in terms of behaviour problems and early signs of attention-deficit/hyperactivity disorder will also be assessed (in the last assessment wave), with the aim of identifying potentially problematic EF trajectories earlier than presently possible.

The Team

We are a small team consisting of Dr Holmboe, her PhD students, research staff, and interns and undergraduate students. In our research we aim to follow open science principles – i.e., we pre-register our study hypotheses and/or planned exploratory analyses, preprint our papers, and share our data sets and other study outputs with the public. Dr Holmboe is a hands-on PI, regularly testing participants herself, and the whole team works closely together to keep up with the year-round testing and data processing that is inevitably part of longitudinal research. We support each other in our day-to-day work, career aspirations and in maintaining a good work-life balance.

Relevant publications

Fiske, A., Collins-Jones, L., de Klerk, C., Lui, K. Y. K., Hendry, A., Greenhalgh, I., Hall, A., Dvergsdal, H., Scerif, G., & Holmboe, K. (2024). The Neural Correlates of Response Inhibition across the Transition from Infancy to Toddlerhood: An fNIRS study. *Imaging Neuroscience*.
https://doi.org/10.1162/imag_a_00206

Fiske, A., Klerk, C. d., Lui, K. Y. K., Collins-Jones, L., Hendry, A., Greenhalgh, I., Hall, A., Scerif, G., Dvergsdal, H., & Holmboe, K. (2022). The neural correlates of inhibitory control in 10-month-old infants: a functional near-infrared spectroscopy study. *NeuroImage*.
<https://doi.org/10.1016/j.neuroimage.2022.119241>

Hendry, A., Greenhalgh, I., Bailey, R., Fiske, A., Dvergsdal, H., & Holmboe, K. (2022). Development of directed global inhibition, competitive inhibition and behavioural inhibition during the transition between infancy and toddlerhood. *Developmental Science*, 25(5), e13193.

<https://doi.org/10.1111/desc.13193>

Holmboe, K., Larkman, C., de Klerk, C., Simpson, A., Bell, M. A., Patton, L., Christodoulou, C., & Dvergsdal, H. (2021). The Early Childhood Inhibitory Touchscreen Task: A new measure of response inhibition in toddlerhood and across the lifespan. *PloS One*, 16(12).

<https://doi.org/10.1371/journal.pone.0260695>

Hendry, A., & Holmboe, K. (2021). Development and validation of the Early Executive Functions Questionnaire: A carer-administered measure of Executive Functions suitable for 9- to 30-month-olds. *Infancy*, 1-30. <https://doi.org/https://doi.org/10.1111/inf.12431>

Braithwaite, E. K., Jones, E. J. H., Johnson, M., & Holmboe, K. (2020). Dynamic modulation of frontal theta power predicts cognitive ability in infancy. *Developmental Cognitive Neuroscience*, 45, 100818.

<https://doi.org/10.1016/j.dcn.2020.100818>

Fiske, A., & Holmboe, K. (2019). Neural substrates of early executive function development.

Developmental Review, 52, 42-62. <https://doi.org/10.1016/j.dr.2019.100866>